

Emerging roles of cations in electrocatalytic reduction of CO₂ and CO

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The electrochemical CO₂ and CO reduction reactions (CO₍₂₎RR) represent a promising strategy for enabling a circular carbon economy. Cations have been shown to have a substantial impact on catalytic activity and selectivity in CO₍₂₎RR; however, the mechanisms in play remain debated. In this Review we explore the multifaceted roles of cations in CO₍₂₎RR, categorizing their effects based on the degree of involvement at the molecular level: as indirect mediators, energetic modulators and participants in electron-transfer processes. At each level, we critically evaluate the key mechanistic hypotheses, supporting evidence and existing contradictions of cation effects. Furthermore, we highlight critical knowledge gaps and potential future research directions, in particular regarding cation effects on specific elementary steps in reaction pathways and catalyst-dependent behaviours, to advance our understanding of the interfacial chemistry and to enable the rational design and optimization of CO₍₂₎RR.

Efficient interconversion between electrical and chemical energy via electrocatalysis constitutes a key pillar in advancing sustainable energy systems¹. Electrocatalytic CO₂ and CO reduction reactions, collectively referred to as CO₍₂₎RR in this work, have emerged as a promising pathway for converting a key greenhouse gas into valuable chemicals and fuels, thereby enabling the circular carbon economy^{2–4}. CO₍₂₎RR occur at the interface between electrode and electrolyte, the composition and structure of which are known to substantially impact the performance of reactions within this microenvironment⁴. The cathodic polarization at potentials relevant to the CO₍₂₎RR drives cation accumulation within the electric double layer (EDL), which is expected to enhance the impact of cations on electrode-mediated reactions^{5,6}.

Following the report of Murata and Hori⁷, numerous studies have demonstrated substantial cation effects on CO₍₂₎RR. The activity of CO₍₂₎RR in the presence of alkali metal cations typically follows the sequence Li⁺ < Na⁺ < K⁺ < Rb⁺ < Cs⁺ (refs. 6–10) and is enhanced at higher cation concentrations^{6,11,12}. However, opposite trends have also been reported in specific systems. For instance, the activity of the CO₂-to-methanol conversion catalysed by cobalt phthalocyanine (CoPc) decreases in the order of Li⁺ > Na⁺ > K⁺ > Rb⁺ > Cs⁺ (ref. 13) and excessively high concentrations of cations inhibit, rather than enhance, the activity^{14,15}. In addition, cation effects show facet dependence, as demonstrated by facet-specific variations in activity^{16,17}. These

exceptions highlight the incompleteness of the current understanding of cation effects.

Beyond the intrinsic reaction activity, cations are closely related to the stability of high-current CO₂RR devices, for example, with flow cell and membrane electrode assembly (MEA) configurations. A key challenge in the long-term continuous operation of CO₂RR devices is the precipitation of (bi)carbonates at the cathode that block pores in the gas diffusion electrode and flow channels, leading to a decline in performance and eventual failure¹⁸. The rate of salt precipitation is highly correlated with the solubility of (bi)carbonates in the sequence Li⁺ < Na⁺ < K⁺ < Rb⁺ < Cs⁺ (ref. 19). Common strategies for mitigating salt precipitation involve the concentration management of interfacial water and OH[−], as discussed previously¹⁹.

In this Review we critically assess the major schools of thought on cation effects in the CO₍₂₎RR. Although cation effects have been the subject of a number of recent review- and perspective-type articles^{5,20–22}, deconvoluting cation effects on the basis of modes of interaction between the cations and the interfacial microenvironment has not yet been a focus. We broadly categorize the roles that cations have in affecting electrocatalytic reactions based on their degree of involvement in electrode-mediated reactions at the molecular level (Fig. 1), that is, as indirect modulators, energetic regulators and reaction participants. As indirect modulators in the microenvironment, cations primarily

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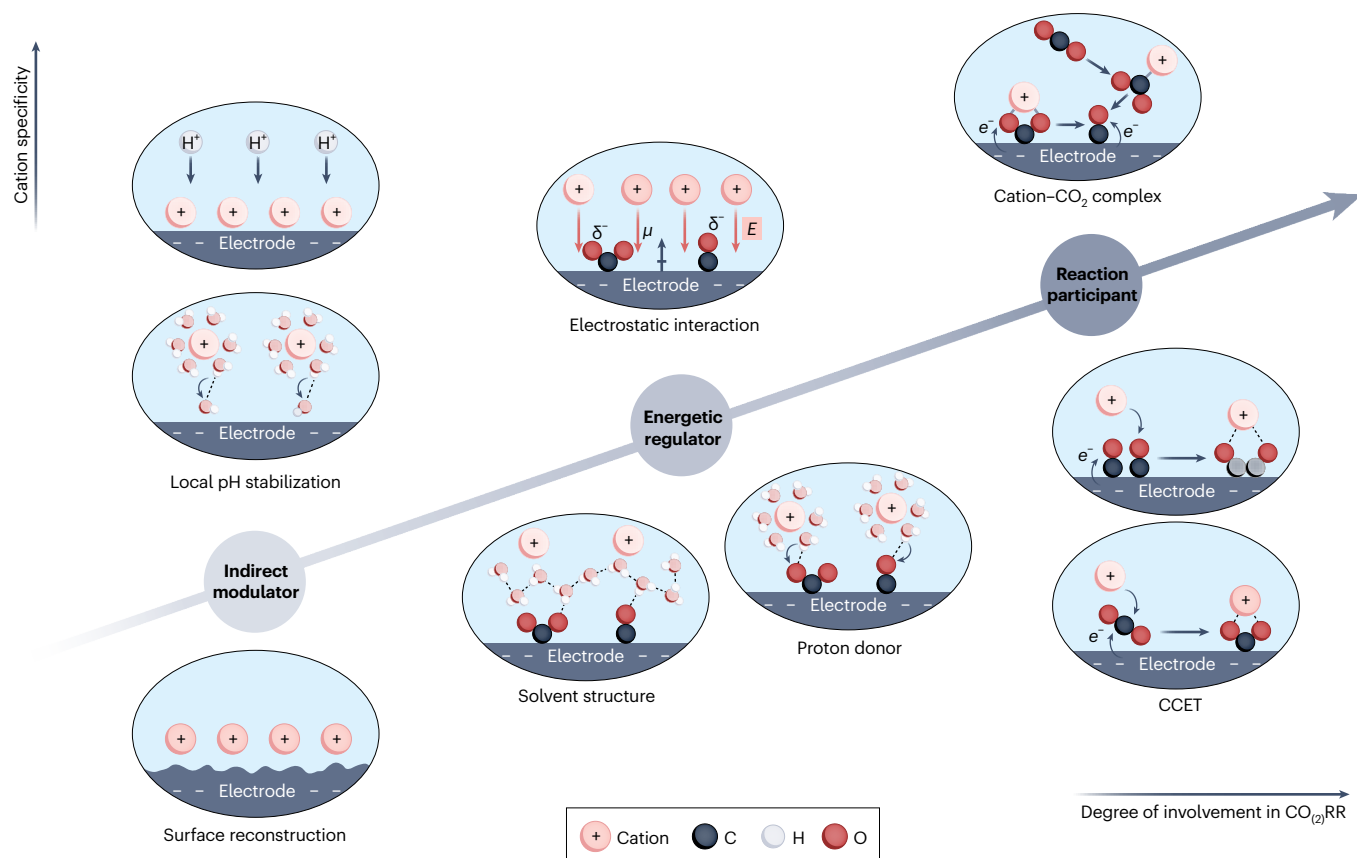


Fig. 1 | Proposed roles of cations in electrocatalysis grouped by the degree of involvement in reactions. In this Review we categorize the roles of cations in CO₂RR into three main aspects, that is, indirect modulator and reaction participant. As indirect modulators, cations affect the local pH and induce surface reconstruction (left). As energetic regulators, cations modify the

energy of intermediates via electrostatic interaction and tune the properties of the solvent and proton donor (middle). As reaction participants, cations serve as a co-catalyst via the formation of a cation–CO₂ complex and coupling with electron transfer (right). δ^- , partial negative charge; μ , dipole moment; E , electric field generated by the charges; e^- , electron.

influence electrocatalytic activity via secondary effects, including interfacial pH modulation and catalyst surface reconstruction, rather than through direct interactions with reactants or intermediates. As energetic regulators of the rate-determining step (RDS), cations modify the energetics and activation barriers of the RDS through electrostatic interactions with intermediates and transition states, as well as via modification of the solvent structure. As participants in elementary reactions, certain metal cations can participate in charge-transfer processes that are analogous to proton-coupled electron transfer (PCET), giving rise to a cation-coupled electron transfer (CCET) mechanism. As the degree of cation involvement increases, cations transition from being part of the microenvironment to an essential component of the reaction, and thus the reaction pathway becomes more cation-specific (Fig. 1). In the following sections, we review the key mechanistic hypotheses and supporting evidence for each role outlined above, identify current knowledge gaps and finally propose promising research directions towards establishing a comprehensive understanding of cation-mediated electrocatalysis.

Cations as indirect modulators in the microenvironment

A significant body of literature indicates that cations neither alter the energetics of intermediates nor directly participate in the elementary steps of CO₂RR but impact the reactivity through indirect mechanisms. One key mechanism involves cation-regulated proton dynamics, where accumulated cations in the EDL influence both proton transport and the local pH via buffering. Another important factor is surface reconstruction, where cation-induced morphological transformations alter the structure,

and in turn the intrinsic activity, of catalytic centres and thus the electrochemical surface area. Typical cation effects are summarized in Table 1, and strategies used to investigate these cation-mediated phenomena are summarized in Box 1 (experimental) and Box 2 (computational).

Interfacial pH

The accumulation of alkali metal cations on the electrolyte side of the EDL close to the negatively charged electrode has been proposed to establish an electrostatic barrier that suppresses proton transport to the electrode surface. In neutral and alkaline electrolytes, the (bi) carbonate produced from the facile reaction of CO₂ with OH[−] decreases the carbon utilization efficiency²³ and obstructs the channels of the gas diffusion electrode via salt precipitation, ultimately compromising the long-term stability of high-current CO₂RR devices¹⁸. This issue can be mitigated in acidic media (pH < 4), where any generated CO₃^{2−} or HCO₃[−] reverts to CO₂ on diffusing from the electrode surface into the bulk electrolyte²⁴. Meanwhile, the hydrogen evolution reaction (HER) is known to occur preferentially over the CO₂RR in acidic media, such that a barrier composed of highly concentrated alkali metal cations in the EDL is needed to suppress H⁺ migration towards the electrode surface to achieve acceptable Faradaic efficiencies of the CO₂RR. This is typically accomplished using an electrolyte with a concentration of alkali metal cations that is several orders of magnitude higher than that of protons^{25–29}. Based on the Gouy–Chapman–Stern (GCS) model, a higher interfacial cation concentration leads to a steeper potential shift across the interface, making the potential at the outer Helmholtz plane (OHP) at the cathode less negative and thus reducing the driving force of H⁺ migration towards the interface^{26,29} (Fig. 2a).

Table 1 | Typical cation effects on the kinetics of CO₂RR (298 K, 1atm)

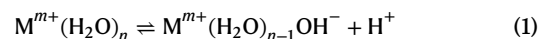
Product	Catalyst	Performance	Potential	$j_{\max}/j_{\min}^a$	Electrolyte	Ref.
HCOOH	Pd/C	Li ⁺ < Na ⁺ < K ⁺ < Rb ⁺ < Cs ⁺	−0.2 V _{RHE}	~2	0.1M MHCO ₃	98
	Cu(111)	Li ⁺ < Na ⁺ < K ⁺ < Rb ⁺ < Cs ⁺	−1.0 V _{RHE}	~7.5	0.1M MHCO ₃	9
	Cu(100)	Li ⁺ < Na ⁺ < K ⁺ ≈ Rb ⁺ ≈ Cs ⁺	−1.0 V _{RHE}	~7.5	0.1M MHCO ₃	9
CO	Poly Au	Li ⁺ < Na ⁺ < K ⁺ < Cs ⁺	−1.0 V _{RHE}	~3	0.1M MHCO ₃	99
	Au(111)	Li ⁺ < Na ⁺ < K ⁺ ≈ Rb ⁺ ≈ Cs ⁺	−0.5 V _{RHE}	~3	0.1M MHCO ₃	16
	Au(100)	Li ⁺ < Na ⁺ < K ⁺ < Rb ⁺ < Cs ⁺	−0.5 V _{RHE}	~28	0.1M MHCO ₃	16
	Au(110)	Li ⁺ < Na ⁺ < K ⁺ < Rb ⁺ ≈ Cs ⁺	−0.5 V _{RHE}	~3	0.1M MHCO ₃	16
	EP-poly Au	Li ⁺ < Na ⁺ < K ⁺ < Rb ⁺ ≈ Cs ⁺	−0.5 V _{RHE}	~7	0.1M MHCO ₃	16
	FA-poly Au	Li ⁺ < Na ⁺ < Cs ⁺ < K ⁺ < Rb ⁺	−0.5 V _{RHE}	~7	0.1M MHCO ₃	16
	Poly Au	TMA ⁺ > TEA ⁺ > TPA ⁺	−0.8 V _{RHE}	~55	0.1M TXAHCO ₃	73
	Ag(111)	Li ⁺ < Na ⁺ < K ⁺ < Cs ⁺	−1.0 V _{SHE}	~15	0.1M MHCO ₃	51
	Sputtered Ag-PTFE	Li ⁺ < Na ⁺ < K ⁺ < Rb ⁺ < Cs ⁺	−1.3 V _{SHE}	~10	0.05M M ₂ CO ₃	87
	Poly Ag	TEA ⁺ > TBA ⁺ > TPA ⁺ > THA ⁺	−2.1 V _{Ag/Ag⁺}	~35	0.5M TXAClO ₄ in CH ₃ CN	52
	Cu(111)	Li ⁺ ≈ Na ⁺ ≈ K ⁺ ≈ Rb ⁺ ≈ Cs ⁺	−1.0 V _{RHE}	~1	0.1M MHCO ₃	9
	Cu(100)	Li ⁺ ≈ Na ⁺ ≈ K ⁺ ≈ Rb ⁺ ≈ Cs ⁺	−1.0 V _{RHE}	~1	0.1M MHCO ₃	9
CH ₄	Cu(111)	Li ⁺ < Na ⁺ ≈ Rb ⁺ ≈ Cs ⁺ < K ⁺	−1.0 V _{RHE}	~1.8	0.1M MHCO ₃	9
	Cu(100)	Li ⁺ ≈ Cs ⁺ < Na ⁺ ≈ Rb ⁺ < K ⁺	−1.0 V _{RHE}	~4	0.1M MHCO ₃	9
CH ₃ OH	CoPc/CNTs	Li ⁺ > Na ⁺ > K ⁺ > Cs ⁺	−0.85 V _{RHE}	~3	0.1M MHCO ₃	13
	CoPc/CNTs	K ⁺ -18-crown-6 < Li ⁺ < K ⁺	−0.94 V _{SHE}	~47	0.1M MHCO ₃	79
C ₂ ^b	Cu(111)	Li ⁺ < Na ⁺ < K ⁺ < Rb ⁺ < Cs ⁺	−1.0 V _{RHE}	~40	0.1M MHCO ₃	9
	Cu(100)	Li ⁺ < Na ⁺ < K ⁺ < Rb ⁺ < Cs ⁺	−1.0 V _{RHE}	~10	0.1M MHCO ₃	9
	Cu foil	TMA ⁺ > TEA ⁺ > TPA ⁺ ≈ TBA ⁺ ≈ 0	−1.29 V _{SHE}	—	0.1M TXAOH	74
	Cu microparticles	Li ⁺ < Na ⁺ < K ⁺ < Cs ⁺	−0.75 V _{RHE}	~4	0.1M MOH	6
	Dendritic Cu	Li ⁺ < Na ⁺ < K ⁺ < Rb ⁺ < Cs ⁺	−0.7 V _{RHE}	~5	0.1M MOH	10
	Sputtered Cu-PTFE	Li ⁺ < Na ⁺ < K ⁺ < Cs ⁺	−1.45 V _{SHE}	~1.7 (flow cell)	0.2M MOH	14
	Sputtered Cu-PTFE	Li ⁺ < Na ⁺ < K ⁺ < Cs ⁺	−1.2 V _{SHE}	~4.5 (H cell)	0.1M MHCO ₃	
	Sputtered Cu-CP	Cs ⁺ < Li ⁺ < K ⁺	−1.4 V _{SHE}	~3	1M MOH	87
			−1.3 V _{RHE}	~50	0.5M M ₂ SO ₄ (pH=2.3)	15

Poly, polycrystalline; EP, electrochemically polished; FA, flame annealed; PTFE, polytetrafluoroethylene; CNTs, carbon nanotubes; CP, carbon paper; M, alkali metal; TMA⁺, tetramethylammonium; TEA⁺, tetraethylammonium; TPA⁺, tetrapropylammonium; TBA⁺, tetrabutylammonium; THA⁺, tetrahexylammonium; TXA⁺, tetraalkylammonium; RHE, reversible hydrogen electrode; SHE, standard hydrogen electrode. ^aAs reaction rates depend on multiple factors, for example, the applied potential, reactor configuration and catalyst structure, comparison among absolute values of current densities from multiple reports obscures the cation effect. Thus, we use the ratio between the maximum and minimum of the partial current densities ($j_{\max}/j_{\min}$) to highlight the cation effect on the kinetics. ^bC₂: C₂H₄, CH₃COOH, C₂H₅OH and *n*-C₃H₇OH.

Beyond free cations, surface-immobilized cationic ionomers have been used as a shielding layer to improve the performance of the CO₂RR in acidic solutions^{30,31} (Fig. 2b), which enables the CO₂RR in a metal-cation-free acidic electrolyte and completely avoids the formation of carbonate precipitates. Analogous to the interfacial concentration of free cations in Fig. 2a, a higher charge density of the ionomer layer (ρ_{ionomer}) leads to a more pronounced suppression of the HER due to stronger electrostatic shielding against H⁺ (Fig. 2b). Similarly, peripheral functionalization of molecular catalysts with hydrophobic quaternary ammonium groups has been shown to be an effective strategy by which to suppress the HER in strong acids^{32,33}. Whereas cations restrict proton migration due to the electrostatic repulsion, they can also hinder CO₂/CO mass transport, especially at very negative potentials with compact EDL structures³⁴, indicating that a balance between reactant mass transport and cation accumulation needs to be achieved to obtain optimal activities^{14,35}.

Continuous proton consumption during CO₂RR is expected to establish a steep pH gradient across the electrochemical interface, with the local pH near the electrode surface substantially exceeding that of

the bulk electrolyte³⁶. This pH disparity becomes more pronounced in high-current-density operations, where rapid proton depletion creates an alkaline microenvironment that affects both the local CO₂ concentration and the thermodynamic favourability of various reaction pathways, thus impacting the product distribution. Hydrated metal cations can be considered as weak proton acids due to their hydrolysis (equation (1)):



When the local pH is higher than the pK_a of hydrated cations, dissociation of water molecules in the hydration layer releases protons, thereby buffering the local pH (Fig. 2c). In bulk electrolytes, the pK_a of alkali metal cations increases slightly with increasing cation size due to the decrease in the electrostatic interaction between cations and the O atoms of water molecules in the hydration shell⁸. Near the cathode, however, the opposite trend of pK_a was proposed after taking the electrostatic interaction between the negative charged surface and cations into account⁸ (Table 2). Thus, the local pH decreases with the

BOX 1

Experimental methods for investigating cation effects

Considering the challenge of directly observing cations at the interface, in particular alkali metal cations, studies of cation effects typically involve observing the responses of a reaction system when varying the identity and concentration of cations. The figure summarizes the experimental methods that have been used for investigating cation effects in the literature.

Potential sweep methods, such as linear sweep voltammetry and cyclic voltammetry, are suitable for investigating the inhibition of cations on H^+ migration²⁶ and the cation-dependent current density (j) values of CO_2 -to- CO conversion experiments when combined with rotating ring-disk electrode measurements^{16,73,99}. Electrochemical impedance spectroscopy is frequently used to determine differential capacitances and surface charge densities with different cations^{50,51,87}. The obtained signals involve multiple processes and require careful deconvolution.

Kinetic measurements of key parameters are summarized in the left panel. Electrode kinetic analysis is an accessible and informative technique for interrogating cation-dependent reaction mechanisms¹⁰³. If cations participate in elementary reactions, it is informative to determine the reaction order (RO) of the cations by varying the cation concentration ($[\text{M}^+]$)⁸⁷. Furthermore, cations may affect key kinetic parameters, that is, the activation energy ($\Delta G^\ddagger$) and electron-transfer coefficient (α), which can be determined from rate

measurements by the varying temperature (T)^{6,99} and the electrode potential (E)^{16,73,99}, respectively.

The right panel exhibits in situ techniques used to characterize the interface. As discussed in the main text, cations can induce surface reconstruction during $\text{CO}_{2\text{RR}}$, which requires in situ spatially resolved characterization techniques, for example, electrochemical scanning probe microscopy (EC-SPM)^{43,86}. Chemical information at interfaces is obtained using vibrational spectroscopy methods, such as ATR-SEIRAS^{6,10,15,34,77}, surface-enhanced Raman spectroscopy^{14,44} and vibrational sum frequency generation (VSFG) spectroscopy^{71,79}, which are widely used in analysing the EDL structure from the signal of probe molecules, for example, adsorbed CO ($^*\text{CO}$) and interfacial water. Some special spectroscopic methods, such as temperature-dependent and high-pressure ATR-SEIRAS, can provide information regarding the effect of cations on CO adsorption thermodynamics⁵ and the rigidity of the EDL³⁴. Time-resolved in situ spectroscopies, such as ultrafast two-dimensional infrared spectroscopy¹⁰⁴, could be promising methods by which to capture transient cation-adsorbate interactions. Furthermore, combining scanning probe microscopy with vibrational spectroscopy can provide spatially resolved chemical information on the electrode surface¹⁰⁵, thus advancing the understanding of cation effects on various surface structures at the nanoscale.

BOX 2

Multiscale simulation of cation effects

Multiscale simulations from ångström- to micrometre-length scales can capture comprehensive pictures of cation effects and assess the relative importance of various effects with different degrees of involvement. In early computational studies with density functional theory (DFT)-based calculations, cation effects were treated as the electric field-dipole interaction, which stabilized the adsorbates of the CO_2RR ^{9,47,48}. A recent report demonstrated that cations changed surface-adsorbate interactions, bringing about changes in the slopes and offsets of adsorption energy scaling relations¹⁰⁶. The effects of cationic surfactants on the adsorption energies of $^*\text{COOH}$ and $^*\text{OCHO}$ on Au, Ag and Zn are different on terrace and defect sites¹⁰⁷. Furthermore, ab initio molecular dynamics (AIMD) can provide insights into the dynamic structure of the EDL and the interactions among cations, adsorbates and the surface, such as cation-specific adsorption⁶⁴ and the coordination between cations and $^*\text{CO}_2^-$ (refs. 54,55,87) or $^*\text{OCCO}^-$ (refs. 59,75,87). Constrained DFT-MD and slow-growth DFT-MD have been used to distinguish the inner- and outer-sphere electron-transfer mechanisms of the CO_2RR ⁵⁸. Beyond typical DFT and AIMD calculations at the ångström scale, cation

effects on the EDL structure and mass transport were investigated via generalized modified Poisson-Nernst-Planck equations with the consideration of volume exclusion due to the size of the solvated cations in the nanometre to micrometre scale⁸³.

Recently, multiphysics models have been developed to simulate cation effects^{35,96,97}, which are beneficial for constructing the complete picture of cation effects by connecting interactions across multiple length scales. Meanwhile, gaps between simulations and experiments remain substantial. Key challenges include the ever-increasing complexity of experimental systems in the pursuit of enhanced activity and selectivity towards specific products, for example, multi-component catalysts, surface modifications and designer electrolytes, the inherent uncertainty in simplified models used in computations, and the accuracy of methods that are capable of taking all interfacial species explicitly into account. Whereas it is impractical to demand that experimental and computational approaches investigate similar systems at comparable scales of length and time in the foreseeable future, it is critical to ensure a certain degree of overlap so that mutual validations can be carried out¹⁰⁸.

sequence of $\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$, as supported by spectroscopic³⁷ and kinetic³⁸ measurements.

The interfacial CO_2 concentration is expected to decrease at higher local pH, which has been proposed as the cause of the increasing CO_2RR activity of $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Rb}^+ < \text{Cs}^+$ (ref. 8). However, the interfacial CO_2 concentration also depends on the consumption rate of CO_2 aside from the local pH. Using in situ attenuated total reflection surface-enhanced infrared absorption spectroscopy (ATR-SEIRAS), direct comparison of interfacial CO_2 concentrations in the presence of different alkali

metal cations shows that the strength of the CO_2 signal follows the sequence $\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$, implying that the interfacial CO_2 concentration largely depends on the rate of CO_2 depletion rather than the cation buffering capacity in a batch cell³⁹. This indicates that the proposed enhanced buffering capacity of larger cations is unlikely to account for the cation effect in the CO_2RR in batch cells, which is even less likely to have much impact on the interfacial pH in devices that are capable of operating at high current density, such as in flow cell and MEA configurations.

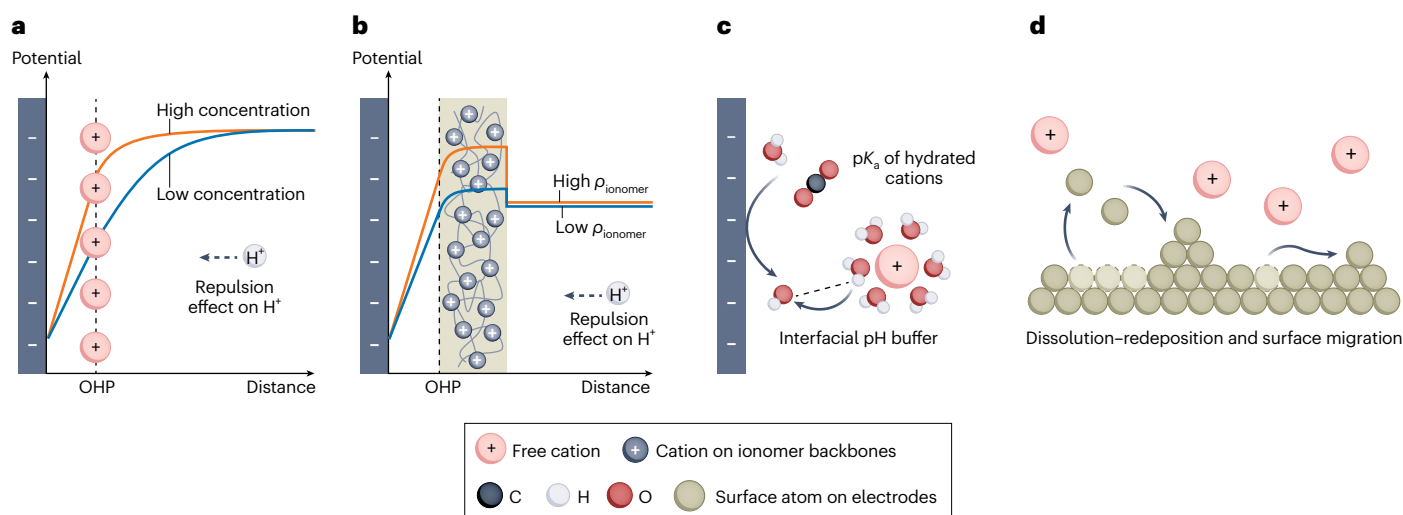


Fig. 2 | Cations as indirect modulators in the microenvironment.

a, Accumulated cations at the OHP hindering H⁺ migration towards the electrode surface. **b**, Surface-immobilized cationic ionomer hindering H⁺ migration towards the electrode surface. **c**, Hydrated cations as the interfacial

pH buffer, determined by the pK_a of hydrated cations at the interface. Hydrated cations serve as the buffer to neutralize the OH[−] generated at the interface, to stabilize the interfacial pH. **d**, Cation-induced surface reconstruction with the mechanisms of dissolution–redeposition and surface migration.

Table 2 | Characteristics of alkali metal cations

	Li ⁺	Na ⁺	K ⁺	Rb ⁺	Cs ⁺	Ref.
Hydration free energy (eV)	−4.95	−3.80	−3.07	−2.86	−2.60	100
Ion radius (Å)	0.69	1.02	1.38	1.49	1.70	100
Hydrated cation radius (Å) ^a	3.37	2.44	—	1.90	1.70	101
	3.82	3.58	3.31	3.29	3.29	102
Effective interfacial cation radius (Å) ^b	5.8	5.2	4.1	3.9	3.5	51
pK _{a, bulk}	13.6	14.2	14.5	14.6	14.7	8
pK _{a, interface} ^c	11.64 (13.16)	10.26 (11.44)	7.95 (8.49)	6.97 (7.23)	4.31 (4.32)	8

^aDetermined via Stokes' law as described in refs. 101 and 102. ^bDetermined using the CO electrochemical Stark shift at Pt and Cu as described in ref. 51. ^cThe pK_a of hydrated cations near the Ag (and Cu, in parentheses) electrode at −1V_{RHE} as described in ref. 8.

Surface reconstruction

Despite the thermodynamic gap between the standard reduction potentials of alkali metals (around −3 V_{SHE})⁴⁰ and typical operating potentials in CO₂RR (no more negative than −2 V_{SHE}), interactions between alkali metal cations and electrode surfaces could be substantial. These interactions, primarily mediated through electrostatic forces and surface polarization effects, could induce the restructuring of catalyst surfaces, that is, cathodic corrosion⁴¹, via a dissolution–redeposition mechanism⁴² (Fig. 2d). Electrochemical scanning tunnelling microscopy has demonstrated that larger alkali metal cations are able to induce more significant surface reconstructions on Au(111)⁴³, altering the density and coordination environment of active sites, thereby modulating both the intrinsic activity and selectivity of CO₂RR. Alkali metal cations have also been reported to promote the dissolution–redeposition of Cu during pulse electrolysis with the effectiveness following the order Li⁺ < Na⁺ < K⁺ < Cs⁺, which impacts the activity by altering the surface Cu_xO/Cu ratio⁴⁴. Reconstructed Cu has been reported to promote alkali metal cation adsorption onto Cu, which is beneficial for the formation of CO₂RR intermediates⁴⁵. In addition to metals that are capable of catalysing CO₂RR, for example, Au and Cu, cation-dependent cathodic corrosion is widely observed on platinum group metals⁴⁶. These reports indicate that cation-induced surface reconstructions could also be present in many other electrocatalytic systems.

When cations act as indirect modulators, they impact the microenvironment in which electrocatalytic reactions take place, for example, the proton availability and surface structure, rather than

interact with the species involved in the reactions, that is, they are essentially spectators.

Cations as energetic regulators of the rate-determining step

The promotional effect of cations on CO₂RR has been widely attributed to regulating the energy barrier of the RDS by activating substrates and stabilizing intermediates and activated complexes (Fig. 3a). Cations primarily interact with reaction intermediates and activated complexes via electrostatics and the interfacial water network. In addition, cations have been proposed to modulate the proton-transfer kinetics by altering the activity of proton donors, especially in neutral and alkaline aqueous electrolytes, thereby influencing the reactivity of hydrogenation pathways.

Electrostatic interaction

There are two general interpretations regarding the electrostatic interaction of cations with reaction intermediates and activated complexes. One proposes that the interfacial electric field affects the energy of intermediates and activated complexes, which is a 'mean field' mechanism that only takes the collective behaviour of the cations into account. The other interpretation considers specific interactions between cations and species in the EDL.

The intense interfacial electric field across the EDL is widely considered as a driver of electrochemical reactions. To a first approximation, the EDL can be viewed as a planar capacitor, with a negatively charged

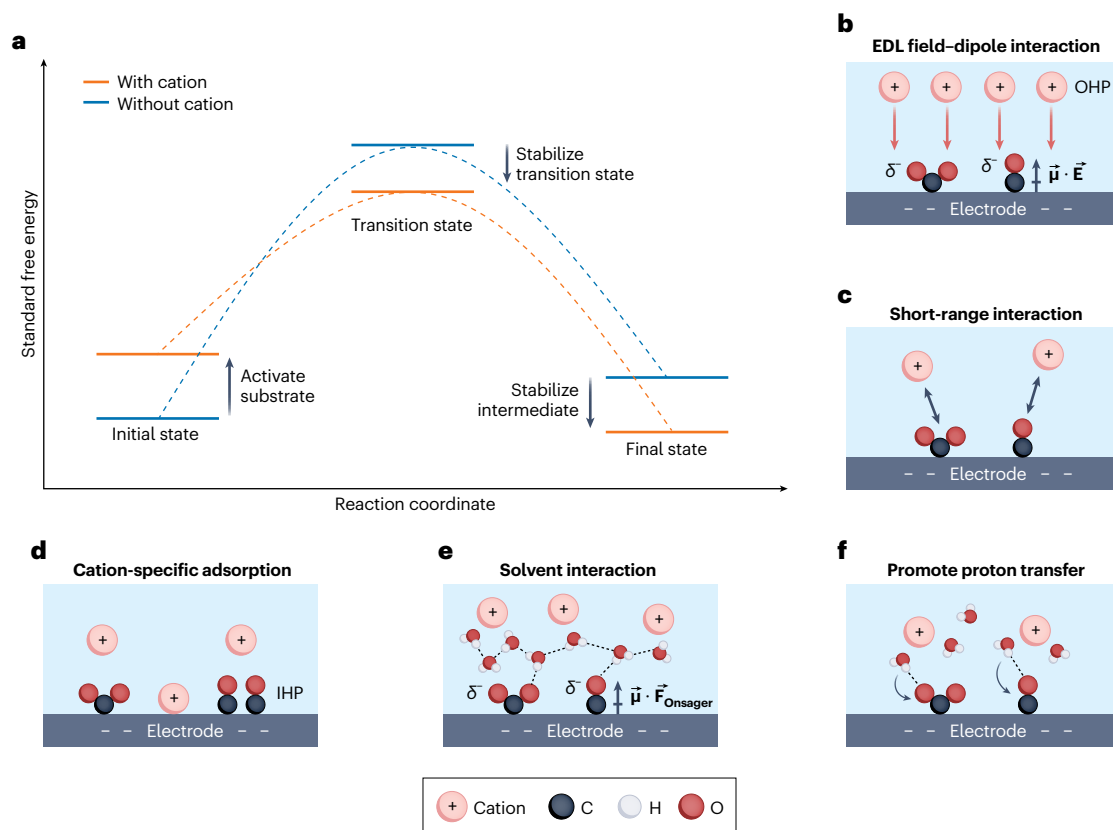


Fig. 3 | Cations as energetic regulators of the RDS. **a**, Effect of cations on the free energy barrier of the RDS. **b**, Interaction between the local electric field and adsorbed dipoles. The electrostatic effect of cations is considered as a mean field and is primarily determined by the charge and radius of the cations. $\vec{\mu}$, dipole moment of adsorbates; $\vec{E}$, local electric field. **c**, Short-range interaction between cations and adsorbates. Although alkali metal cations and intermediates of CO_2 RR are not typical coordination centre atoms and ligands, respectively, the quasi-coordination or Lewis acid–base interaction has been proposed between cations and intermediates in CO_2 RR systems. **d**, Co-adsorption of cations

and $^*\text{CO}_2/^*\text{CO}$. Owing to the strong polarization of cations, the electrons of surrounding species could be redistributed, thereby affecting the adsorbates in the surroundings. IHP, inner Helmholtz plane. **e**, Cation-dependent solvent interaction with adsorbates. Cations indirectly affect the energy of intermediates by tuning the solvent structure, which determines the hydrogen bonding and the strength of the Onsager reaction field (F_{Onsager}). **f**, Cations promoting water dissociation. Cations with a high charge density can polarize the bonds of surrounding water molecules, thus activating interfacial water and promoting proton transfer.

electrode and an OHP dominated by cations in cathodic reactions. Interfacial species are exposed to the intense electric field across the EDL ($\sim 10^8$ – 10^9 V m^{-1})^{47–49}, so that the stability of reaction intermediates and activated complexes, especially species with (partial) charges and large dipole moments, is expected to be impacted by the magnitude of the interfacial electric field^{9,48} (Fig. 3b). The OHP is considered to be the distance of closest approach of non-specifically adsorbed ionic species⁵⁰, so that the distance between the electrode and the OHP is expected to mirror the size of hydrated cations. It follows that the interfacial electric field strength is inversely proportional to the size of the hydrated cations at a given electrode potential (assuming that the potential of zero charge, or PZC, does not vary with the cations), giving rise to varying degrees of electrostatic stabilization of intermediates and activated complexes.

The size of hydrated alkali cations follows the order $\text{Li}^+ > \text{Na}^+ > \text{K}^+ \geq \text{Rb}^+ \geq \text{Cs}^+$ (refs. 10,51) (Table 2), leading to the strongest stabilization effect on adsorbed dipoles in the presence of Cs^+ among alkali metal cations⁶. This interfacial electric field has been considered to be the main driver in the non-aqueous CO_2 RR system with small organic alkylammonium cations⁵². Based on the GCS model, higher interfacial concentrations of cations result in the stronger local electric field with more pronounced stabilization effects on the intermediates¹². The electrostatic stabilization of reaction intermediates offers valuable insights for molecular catalyst design. By incorporating peripheral functional groups, such as quaternary ammonium groups, the local electric field

around the active sites can be precisely modulated, thereby regulating the catalytic activity^{32,33,53}. It is worth noting that even with a comparable interfacial electric field strength, as determined by the Stark tuning rates, different cations still lead to distinct CO_2 RR activities. These results imply the importance of non-electric field components of the cation effect¹⁰, which will be discussed below.

Aside from the mean-field interpretation, cations have been proposed to interact specifically with surface-bound reaction intermediates and activated complexes via short-range Coulombic attraction (Fig. 3c) and specific adsorption on the electrode surface (Fig. 3d). Hydrated alkali metal cations with a more loosely coordinated (or softer) hydration shell are smaller in size (Table 2), for example, Cs^+ , so that they can be packed more densely at the interface. Higher concentrations of interfacial cations are more likely to stabilize $^*\text{CO}_2^-$ in CO_2 -to- CO conversion^{54–58}, resulting in the activity trend of $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Rb}^+ < \text{Cs}^+$. In addition, CO adsorption thermodynamics and activation parameters of the RDS have been experimentally determined⁶, showing that CO adsorption becomes less favourable and the activation barrier of the RDS is lower in the presence of alkali metal cations with a softer hydration shell. Taking Cs^+ as an example, it achieves the lowest energy barrier for the RDS among alkali metal cations by raising the free energy of $^*\text{CO}$ while simultaneously stabilizing the transition state via electrostatic interactions. AIMD simulations indicate that the propensity of weakly hydrated cations to form short-range interactions with $^*\text{CO}$ could facilitate its activation. These results imply that the interaction

between alkali metal cations and $^*\text{CO}$ effectively activates the initial state and stabilizes the activated complex in the RDS⁵⁹ (Fig. 3a). In this regard, a Marcus-type model is developed with the consideration of the electrostatic interaction between cations and negatively charged intermediates, providing a theoretical framework by which to rationalize how cations stabilize negatively charged intermediates⁶⁰.

Computational simulations provide a molecular-level picture of cation effects on the specific mechanism, indicating that the electrochemical activation of CO_2 on Au proceeds via an inner-sphere electron-transfer pathway to a $^*\text{CO}_2\text{-M}^+$ complex (M^+ = alkali metal cation)⁵⁷, whereas CO_2^- produced via an outer-sphere electron-transfer pathway cannot adsorb on Au without alkali metal cations⁵⁸. For the CO_2RR catalysed by Ni single-atom catalysts, the RDS was proposed to be a PCET step ($^*\text{CO}_2 + \text{H}^+ + e^- \rightarrow ^*\text{COOH}$) in acidic solutions without K^+ , which was shifted to a proton-transfer step ($^*\text{CO}_2 \cdots \text{K}^+ + \text{H}^+ \rightarrow ^*\text{COOH} + \text{K}^+$) due to the reduced energy barrier of the first electron-transfer step in the CO_2RR via the stabilization of K^+ on $^*\text{CO}_2^-$ (ref. 61). In addition, the more pronounced stabilization effect of cations on adsorbates containing C–C bonds relative to C_1 intermediates is consistent with the calculated promotional effect of cations for C_2 formation on Cu electrodes¹⁷. In some cases, short-range interactions between metal cations and adsorbates are interpreted as the direct participation of cations in reactions, which will be discussed in the following section.

Although standard reduction potentials of alkali metal cations are much more negative than typical potentials used in $\text{CO}_{(2)}\text{RR}$, several reports have indicated that alkali cations can be specifically adsorbed on the electrode with partial charge transfers^{62–64}. Adsorbed cations are expected to modify the local charge distribution, which promotes C–C coupling, especially with the co-adsorption of cations and CO (ref. 64). However, via *in situ* SEIRAS experiments, an excessively high Cs^+ concentration was observed to shift the preference of $^*\text{CO}$ on the Cu surface from the atop configuration to the bridge configuration, leading to a decline in C–C coupling, which was attributed to the altered electronic environment due to Cs^+ adsorption¹⁵. It should be noted that, if specific adsorption refers only to species forming a covalent bond with the surface, additional evidence is needed to support the specific adsorption of cations on electrode surfaces at potentials that are relevant for $\text{CO}_{(2)}\text{RR}$.

Solvent structure

Solvent molecules are the most abundant species at electrochemical interfaces; these interact extensively with all other interfacial species, and thus are expected to exert significant energy-modulating effects on reactions. In the case of aqueous electrolytes, ions at the interface interact with water via hydration, and water molecules in the hydration shells of ions interact with other interfacial water molecules via hydrogen bonding. Thus, the nature and concentration of accumulated cations via cathodic polarization close to the electrode are expected to shape the interfacial water structure^{65,66}, which in turn will affect the interaction between interfacial water and reaction intermediates (Fig. 3e). According to the Onsager solvation theory⁶⁷, polarized solvent molecules create a field in the cavity occupied by the solute, which is known as the Onsager reaction field⁶⁸. The dielectric properties of the interfacial solvent environment are significantly different from those in the bulk solution and vary with cations at the interface^{69,70}, resulting in the cation-dependent strength of the Onsager reaction field. The Stark tuning rates measured via plasmon-enhanced VSFG spectroscopy demonstrate that the Onsager reaction field with alkali metal cations follows $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Cs}^+ < \text{Rb}^+$, which indicates that the Onsager reaction field needs to be explicitly considered for CO_2RR kinetics⁷¹. It should be noted that, although both the interfacial electric field and the Onsager reaction field are ultimately driven by the electrode potential, the origins of the immediate source of the fields, that is, the electrode and solvent molecules, are different.

The solvent effect caused by cations has been correlated with certain kinetic parameters, such as the activation entropy in the Eyring equation⁷² and the reorganization energy in Marcus kinetics⁷³, enabling a quantitative description of the solvent effect on $\text{CO}_{(2)}\text{RR}$ kinetics. A recent study has demonstrated that high NaClO_4 concentrations increase the apparent activation entropy due to the disorder of the interfacial water structure, leading to an 18-fold enhancement in the current density of C_2 products⁷². Furthermore, larger alkali metal cations, such as Cs^+ , are able to decrease the reorganization energy of the CO_2RR due to the breaking effect on the hydrogen-bond network⁷³. Bulkier organic cations increase the fraction of non-polar components and lower the interfacial water concentrations, resulting in lower polarity, that is, a smaller static dielectric constant at the interface⁷³. Based on the Born solvation model⁵⁰, smaller static dielectric constants lead to lower outer reorganization energies, and this is considered to be the main cause for the lower measured reorganization energy in bulkier organic cations.

Interfacial water has been proposed to interact with negatively charged intermediates via hydrogen bonding in aqueous electrolytes^{74–77}, which could stabilize the intermediate of CO–CO coupling, that is, $^*\text{OCCO}$ (refs. 74,75). Larger organic cations disrupt the water structure, hindering the interaction between surface-bound species and interfacial water molecules, inhibiting the formation of C_2 products⁷⁴. By contrast, in the presence of alkali metal cations, the interaction between $^*\text{CO}_2/^*\text{CO}$ and the hydrogen in interfacial water becomes beneficial to the hydrogenation reaction, leading to C_1 rather than C_2 products⁷⁵. From this perspective, larger alkali metal cations with structure-breaking effects on interfacial water⁷⁸ approach the interface through desolvation and coordinate with the $^*\text{CO}$, partially substituting the hydrogen-bond stabilizing effect of interfacial water, resulting in higher C_2 selectivity⁷⁵. The contrasting impacts of alkali metal and organic cations on hydrogen bonding and C_2 selectivity in $\text{CO}_{(2)}\text{RR}$ could be attributed to different modes of interaction between cations and adsorbates. With alkali metal cations, both cations and interfacial water have been proposed to stabilize reaction intermediates^{74,75}. Although larger alkali metal cations tend to weaken the interaction between interfacial water and adsorbed species, the enhanced stabilizing effect could more than compensate for the adverse effects of losing a fraction of the interfacial water⁷⁵. With organic cations, the interfacial water is the only species able to engage in short-range interactions with adsorbates, such that suppressing water-adsorbate interactions with larger organic cations due to their higher hydrophobicity tends to make C–C coupling in $\text{CO}_{(2)}\text{RR}$ more unfavourable⁷⁴.

Proton donor

Water molecules (or hydronium ions in acidic media) serve as the main proton donors in $\text{CO}_{(2)}\text{RR}$, with their interfacial behaviour being modulated by cation-specific effects. Whereas the activity of the proton donor is directly related to the proton-transfer kinetics, the kinetics depend not only on the concentration but also on the microenvironment of interfacial water, for example, its interactions with cations. The local concentration, molecular orientation and hydrogen-bond network of proton sources show pronounced cation dependence, establishing cations as key regulators of proton-transfer kinetics (Fig. 3f). Whether the activity of the proton donor matters depends on whether the RDS includes proton transfer.

For CO_2 -to-CO conversion in acidic media, the PCET of $^*\text{CO}_2$ hydrogenation is considered to be kinetically relevant⁷⁷, where the accessibility of hydrogen atoms in interfacial water and the oxygen atom of CO_2 are the key factor for the reduction kinetics of $^*\text{CO}_2$ (ref. 77). A similar line of reasoning is adopted to rationalize the activity trend in the CO_2 -to-methanol conversion on CoPc ($\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Cs}^+$) with the RDS being a PCET step¹³. Owing to the structure-breaking effect on hydrogen-bonded networks that follows the order $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Cs}^+$ (ref. 78), smaller cations provide a greater availability of interfacial

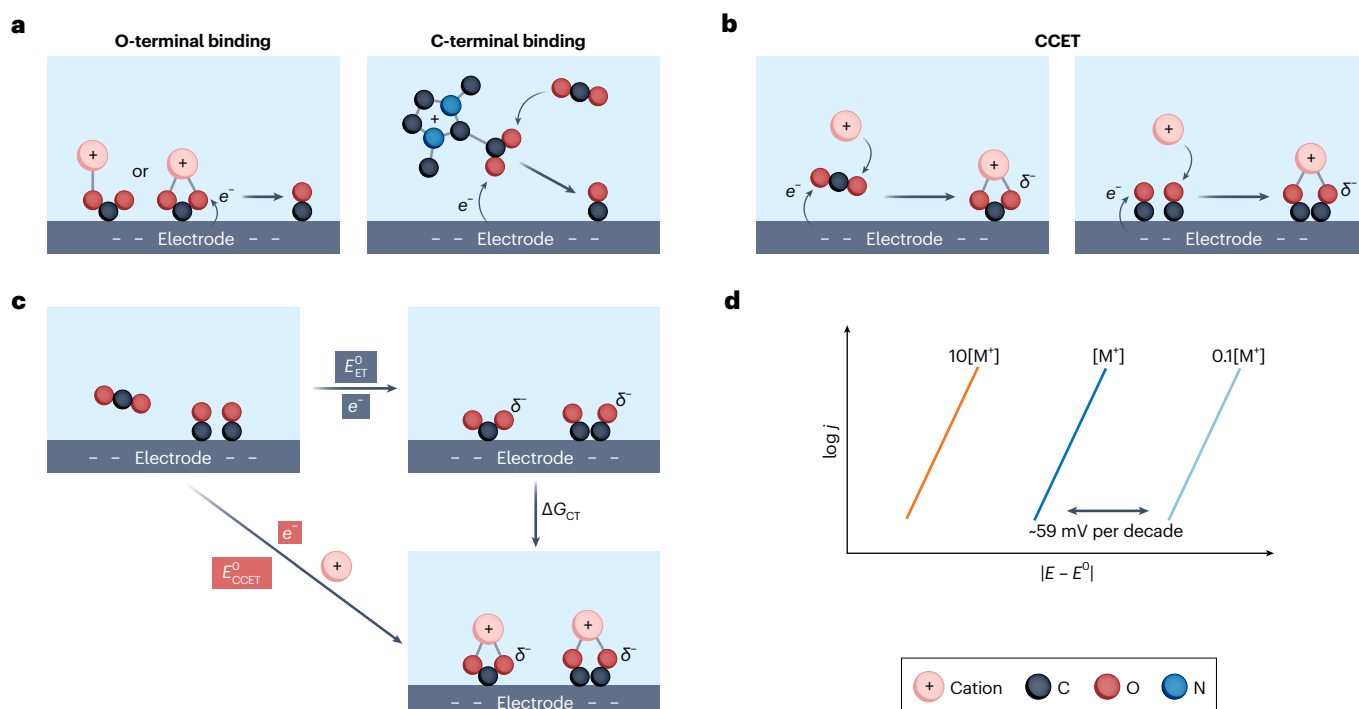


Fig. 4 | Cations as participants in elementary reactions with electron transfers.

a, Cation–CO₂ complexes participating in the reaction network. O-terminal binding, the coordination between cations and activated CO₂ is considered as the Lewis acid–base interaction, facilitating the subsequent reaction of CO₂ (left). C-terminal binding, the N-heterocyclic carbene formed from the imidazolium ring is able to bind with the C in CO₂ (right). Hydrogen atoms omitted for clarity. **b**, CCET in CO₂ adsorption (left) and C–C coupling (right). Cations serve as reactants in the electron-transfer step, forming cation–adsorbate complexes as products. **c**, Reaction scheme of CO₂ adsorption and CO–CO coupling with

cation participation. The thermodynamics of CCET can be described as the sum of sequential electron transfer and cation transfer, which could help to identify contributing factors in CCET. **d**, For the reaction scheme in **b**, the kinetics of CCET in the CO₂RR are expected to produce typical $\log j$ versus E curves with the characteristics shown. Higher cation concentrations are beneficial to the apparent kinetics. The curve is predicted to shift by -59 mV per order of magnitude change in the cation concentration (the assumptions for this are discussed in the main text).

hydrogen bonds near the key intermediate of methanol formation (*CHO). Furthermore, smaller acidic cations, for example, Li⁺, with a high hydration energy, that is, strong cation–water interactions, can polarize water molecules and weaken the O–H bond, promoting proton transfer to *CHO (ref. 13). As the acidity of hydrated cations (that is, the pK_a) at the interface could be different from that in the bulk solutions⁸ (Table 2), the polarization of water by the electrodes also needs to be taken into account. The kinetics of electron transfer are also important in PCET steps aside from proton transfer, which could be the reason for K⁺ being proposed as a better choice for the CoPc-catalysed formation of methanol in the CO₂RR rather than Li⁺ (ref. 79).

Whereas the formation rates of methane^{80,81} and methanol¹³ are probably controlled by PCET steps, the RDS in the formation of C₂₊ products is widely believed to be C–C coupling without proton transfer. Thus, the dependence of the intrinsic kinetics of C₂₊ products on the activity of the proton donor is expected to be weaker than that of C₁ products and the HER. It follows that systems with a lower H₂O activity, such as water-in-salt electrolytes⁸², have the potential to enhance the selectivity of C₂₊ products. With the interfacial concentration of alkali metal cations following the order Li⁺ < Na⁺ < K⁺ < Cs⁺ at a given bulk cation concentration^{6,54,83}, the concentration of interfacial water is expected to follow the opposite trend. The trend of interfacial water concentration has been correlated with C₂₊ selectivity in the presence of different alkali metal cations^{75,82}.

When cations are considered as energetic modulators, they are an integral part of the interactive microenvironment of reactions; however, they do not appear in the formula of any elementary step of the reaction network. Thus, they remain in the microenvironment where reactions take place, rather than being part of the reactions.

Cations as participants in electron-transfer steps

There is an emerging recognition of cations as active participants in CO₂RR rather than as part of the microenvironment in which the reactions take place. As Lewis acids, metal cations have been proposed to coordinate with *CO₂ to form a cation–CO₂[−] complex^{84–87}, which facilitates subsequent reduction reactions. Organic cations, in particular imidazolium cations and their derivatives, have been shown to form stable CO₂ adducts through specific interactions at their C2 positions^{88–90}. More recent research has revealed that certain metal cations can engage in CCET, which is analogous to PCET, expanding our understanding of electrochemical CO₂(₂) conversion pathways.

Cation–CO₂ complex

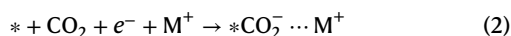
It has been reported that metal cations coordinate directly with surface-adsorbed species, forming cation–adsorbate complexes that participate in subsequent reduction processes (Fig. 4a, left). In early literature accounts of metal complexes, such as Fe(0) porphyrins, catalysing the CO₂RR in non-aqueous electrolytes, it was proposed that metal cations coordinate with activated CO₂ as a Lewis acid to enhance deoxygenation of the Fe(II)–CO₂^{2−} complex to form CO and carbonate^{84,85}. Lewis acidic cations bind with the O terminus of activated CO₂ to stabilize the intermediate and promote O transfer rather than facilitating the electron-transfer step. The Co(I)Pc–CO₂–Mg²⁺ complex in aqueous solution has recently been observed using in situ electrochemical scanning tunnelling microscopy, providing strong experimental evidence for the presence of cation–CO₂ complexes⁸⁶. Unlike molecular catalysts, the structure of the active site on a metal catalyst is typically not as well-defined, making the identification of adsorbed complexes challenging. Whereas direct experimental evidence of

cation–CO₂ complexes on metal catalysts remains scarce, computational results indicate the existence of M⁺–CO₂[−] (refs. 54,55,57,87) and M⁺–OCCO[−] (refs. 59,75,87) species.

Imidazolium cations and derivatives can serve as a Lewis base to form a chemical bond with CO₂. As the proton at the C2 position of the imidazolium ring is acidic, it interacts strongly with the base via hydrogen bonding. Dissociation of the proton at the C2 position leads to an N-heterocyclic carbene, which can form a complex with CO₂ via C-terminal binding⁸⁸ (Fig. 4a, right). The reaction pathway involving the intermediate of the imidazolium–CO₂ complex could significantly lower the total energy barrier of the CO₂RR, resulting in a high conversion efficiency at low overpotential^{89,90}. Building on this insight, the 1-methylbenzimidazolium cation was designed as the fixed charge in the ionomer to conduct ions as well as activate CO₂, and was utilized to realize the co-electrolysis of CO₂/pure water for the production of C₂H₄ at an industrial-scale current density in the MEA configuration⁹¹.

Cation-coupled electron transfer

The involvement of cations as reactants in electron-transfer steps necessitates their explicit incorporation into the thermodynamic analysis of CO₂RR (Fig. 4b). When cations engage in elementary steps through CCET mechanisms, they fundamentally alter both the energetics and the kinetics of the reaction. CO₂ adsorption^{54,87} and CO–CO coupling⁸⁷ have been considered to proceed via the CCET mechanism, as in:



The thermodynamic diagram is shown in Fig. 4c, and the standard potential of the CCET step (E_{CCET}^0) can be expressed as:

$$E_{\text{CCET}}^0 = E_{\text{ET}}^0 - \frac{\Delta G_{\text{CT}}}{F}, \quad (4)$$

where E_{ET}^0 and ΔG_{CT} are the standard potential of the electron-transfer step and the free energy of the cation-transfer step, respectively, and F is the Faraday constant. As the coordination of $* \text{CO}_2^- / * \text{OCCO}^-$ with hydrated M⁺ necessarily involves disruption of the hydration shell of M⁺, ΔG_{CT} is expected to have a contribution from the partial desolvation of M⁺, as in equation (5):

$$\Delta G_{\text{CT}} = \Delta G_{\text{c}} - \chi \Delta G_{\text{hyd}}, \quad (5)$$

where ΔG_{c} and ΔG_{hyd} are the coordination free energy between $* \text{CO}_2^- / * \text{OCCO}^-$ and M⁺ and the hydration free energy of M⁺, respectively, and χ is the coefficient representing the degree of partial dehydration ($0 \leq \chi \leq 1$). Combining equations (4) and (5), E_{CCET}^0 can be expressed as:

$$E_{\text{CCET}}^0 = E_{\text{ET}}^0 + \frac{\chi \Delta G_{\text{hyd}} - \Delta G_{\text{c}}}{F}. \quad (6)$$

It is clear from equation (6) that the thermodynamics of CCET depend on both the energetics of electron transfer and the chemical properties of cations and intermediates.

The kinetics of CCET require the explicit term of the cation's contribution in the rate equation. With the participation of cations, the current density can be described via Butler–Volmer kinetics as:

$$j = n F k^0 [S]_{\text{int}}^{\text{RO}} [M^+]_{\text{int}} \exp \left(- \frac{\alpha F (E - E_{\text{CCET}}^0)}{RT} \right), \quad (7)$$

where $[S]_{\text{int}}$ and $[M^+]_{\text{int}}$ are the interfacial concentrations (activities) of the substrate and cation, respectively, RO is the reaction order of the

substrate, n is the number of electrons transferred and α is the transfer coefficient. The standard rate constant, k^0 , can be expressed as $k^0 = A \exp \left(- \frac{\Delta G_{\ddagger}^0}{RT} \right)$, where A and $\Delta G_{\ddagger}^0$ are the pre-exponential factor and the activation free energy at standard potential, respectively. Based on the GCS model, $[M^+]_{\text{int}}$ can be replaced by the bulk concentration of cations ($[M^+]_{\text{bulk}}$) via $[M^+]_{\text{int}} = \beta [M^+]_{\text{bulk}}^{0.5}$, where β is a proportionality coefficient⁸⁷. With the assumption of $\alpha = 0.5$, equation (7) can be rewritten as:

$$j = n F k^0 [S]_{\text{int}}^{\text{RO}} \beta \exp \left(- \frac{\alpha F (E - E_{\text{CCET}}^0 - 0.059 \log [M^+]_{\text{bulk}})}{RT} \right). \quad (8)$$

Equation (8) predicts that the curve of $\log j$ versus E will shift by around 59 mV per order of magnitude change in the cation concentration (Fig. 4d). This prediction has been verified in the conversion of CO₂ to CO and in the formation of C₂H₄ formation via CO₂RR on Au and Cu, respectively⁸⁷, confirming the direct participation of alkali metal cations in the RDSs of these reactions (Fig. 4b). Similar potential shifts have also been observed in the oxygen reduction reaction on carbon⁹² and in the desorption of sulfonate in Nafion on Pt(111)⁹³, indicating that CCET can be a general mechanism in electrocatalysis.

It is worth noting that an implicit assumption in equation (8) is that both k^0 and E_{CCET}^0 are constants, so that the potential shift at a given j value can be attributed exclusively to the change in the cation concentration. This assumption is reasonable when the identity of the cation remains unchanged⁸⁷, and will probably break down if activities are measured with different cations. It is likely that the energetics of the coordination between cations and intermediates, that is, ΔG_{c} , χ and ΔG_{hyd} (equation (5)), depend on the nature of the cation, so that E_{CCET}^0 and the intrinsic energy barrier ($\Delta G_{\ddagger}^0$) are also cation-dependent. If we denote the difference in activation free energy caused by cation M⁺ relative to a reference state to be $\Delta G_{\text{M}}^{\ddagger,0}$, the cation-dependent standard rate constant, $k^0(\text{M}^+)$, can be expressed as:

$$k^0(\text{M}^+) = k^0 \exp \left(- \frac{\Delta G_{\text{M}}^{\ddagger,0}}{RT} \right). \quad (9)$$

Taking into account the impact of cations on both the thermodynamics and kinetics of CCET, equation (8) can be rewritten by combining equations (6) and (9) as:

$$j = n F k^0 [S]_{\text{int}}^{\text{RO}} \beta \exp \left(- \frac{\alpha F (E - E_{\text{ET}}^0 - \Delta E)}{RT} \right) \quad (10)$$

$$\Delta E = \frac{\chi \Delta G_{\text{hyd}} - \Delta G_{\text{c}}}{F} - \frac{\Delta G_{\text{M}}^{\ddagger,0}}{\alpha F} + 0.059 \log [M^+]_{\text{bulk}}. \quad (11)$$

Equations (10) and (11) show that the potential shift (ΔE) at a given j shows dependence on both the concentration and the identity of the cation.

Beyond the empirical Butler–Volmer framework, the Marcus kinetics have also been used to describe CCET in recent years, and are able to provide more molecular-level insights. A general theory has been proposed to describe coupled ion–electron transfer, which unifies the Marcus kinetics of electron transfer with the Butler–Volmer kinetics of ion transfer⁹⁴. According to the treatment previously, the rate constant of CCET (k_{CCET}) can be expressed as⁹⁴:

$$k_{\text{CCET}} = \frac{\exp \left(- \frac{\Delta G_{\text{CT}}^{\ddagger}}{k_B T} \right)}{\gamma_{\ddagger}^{\text{CT}}} k_{\text{red/ox}}^{\text{MHC}} \quad (12)$$

$$k_{\text{red/ox}}^{\text{MHC}} = \frac{2\pi\hbar^2}{\hbar\sqrt{4\pi\lambda k_{\text{B}}T}} \int_{-\infty}^{\infty} \exp\left[-\frac{(-\varepsilon \pm e(E - E_{\text{ET}}^0) + \lambda)^2}{4\lambda k_{\text{B}}T}\right] \frac{\rho d\varepsilon}{1 + \exp\left(\frac{\pm e\varepsilon}{k_{\text{B}}T}\right)}, \quad (13)$$

where $k_{\text{red/ox}}^{\text{MHC}}$ is the rate constant based on the Marcus–Hush–Chidsey theory for non-adiabatic electron transfer. The variables λ and H represent the reorganization energy and the electronic coupling matrix element, respectively, $\Delta G_{\text{CT}}^\ddagger$ is the free energy barrier of cation transfer, $\gamma_{\text{CT}}^\ddagger$ represents the activity coefficient of the cation-transfer transition state, k_{B} is the Boltzmann constant, $\hbar$ is the reduced Planck constant, e is the elementary charge, ε is the energy relative to the Fermi level and ρ is the density of electronic states. The formulation of equations (12) and (13) requires $\Delta G_{\text{CT}}^\ddagger \ll \lambda$ and non-adiabatic electron transfer⁹⁴. This formalism has been successful in describing lithium intercalation in lithium iron phosphate^{94,95} and CO partial current densities of the Ag-catalysed CO₂RR^{96,97}. Recently, a mechanistic framework based on the Marcus–Hush–Chidsey theory has been developed to explain how both alkali metal and organic cations modulate the thermodynamics and kinetics of the CO₂RR, where the RDS of the Au-catalysed CO₂RR is considered to be CCET with alkali metal cations and electron transfer with organic cations⁷³. The impact of alkali metal and organic cations on the rate of the CO₂RR is limited by different factors. Alkali metal cations are able to maintain large capacitances, that is, surface charge densities, whereas cation transfer introduces an additional energy barrier due to the partial dehydration or distortion of hydrated structures. Organic cations can substantially reduce the reorganization energy due to improved hydrophobicity at the interface but are limited by low surface charge densities to enable efficient electron transfer.

Cations essentially function as co-catalysts participating in electron transfer as a reactant as they will be released in a subsequent step to yield cation-free products in CO₂RR. The signature evidence that supports CCET is the quantitative correlation between the applied potential and the cation concentration at a given (partial) current density.

Outlook

Whereas significant progress has been made in explaining the cation effect on CO₂RR and electrocatalytic reactions in general, the understanding of the complex interplay that exists between cations and dynamic electrochemical interfaces remains patchy at best. Although this Review artificially categorizes cation effects into three broad groups based on the degree of the involvement in CO₂RR (Fig. 1), the apparent cation effects are the overall results across multiple length and time scales. Dominant mechanisms of the cation effect can vary under different conditions. In the CO₂RR in an acidic electrolyte, the primary limitation is the intense competing HER, making the cation-induced inhibition of H⁺ migration the main contribution for selectivity control. In neutral and alkaline electrolytes, a number of pathways have been proposed to be the main contributing factor. It has been suggested that the impact of cations on the surface charge density outweighs that of the solvent reorganization energy, leading to a superior performance with alkali metal cations over organic cations⁷³. A multiphysics model developed previously identified electric field-driven solvent effects as the dominant contributor⁹⁷, whereas simulations in a recent study have highlighted the need for optimal cation concentrations to balance kinetics and CO₂ mass transport³⁵. Despite progress in deconvoluting these factors, a systematic analysis remains lacking, which is largely due to challenges in isolating the multitude of variables involved in the CO₂RR. In this regard, multiscale simulations that closely mimic real systems may offer a viable pathway for further investigations (Box 2).

Below, we identify critical knowledge gaps and propose promising research directions that may advance our understanding of these intricate interfacial phenomena and enable the rational design and optimization of CO₂RR systems and beyond.

Cation effects on elementary steps

Elucidating the impact of cations on individual elementary steps offers crucial molecular-level insights into the microkinetics of electrocatalytic reactions. Such a fundamental understanding is particularly valuable for enabling selectivity control in complex reaction networks, for example, steering towards the formation of multi-carbon (C₂₊) products in CO₂RR. Cation effects on the elementary steps of CO adsorption⁶ and the RDS of CO₂RR^{6,73} have been experimentally investigated by integrating kinetic measurements and in situ characteristics. Unfortunately, such methods cannot provide the information of the elementary steps after the RDS due to their high reaction rates and the short lifetimes of intermediates. Computational simulations are currently the most feasible methods. By precisely mapping how cations modulate activation barriers and intermediate stabilization at each step, we can develop targeted strategies for optimizing reaction pathways towards the desired products while suppressing competing reactions.

PCET and CCET mechanisms

PCET mechanisms, including both concerted and sequential electron-transfer–proton-transfer or proton-transfer–electron-transfer pathways, dominate aqueous electrochemical processes, which can be considered as one subcategory of CCET. The fundamental criteria that govern when and how cations other than protons can effectively participate in electron-transfer processes remain poorly understood. As discussed above, the thermodynamics and kinetics of CCET in CO₂RR could potentially be determined by the interaction between metal cations and negatively charged species and the hydration of cations. Clarifying the specific reaction characteristics that enable alternative cations to facilitate charge transfer, including their hydration properties, ionic radii and interfacial behaviour, presents a promising avenue for advancing our understanding of electrochemical reaction mechanisms and developing novel catalytic strategies.

Catalyst-dependent cation effects

The influence of alkali metal cations on the activity of CO₂RR has exhibited a hitherto perplexing dependence on catalytic materials. Whereas a consistent reactivity trend of Li⁺ < Na⁺ < K⁺ < Rb⁺ < Cs⁺ has been observed for most metal-catalysed systems, the reverse trend is demonstrated in the CoPc-catalysed conversion of CO₂ to methanol¹³. Divergent trends of cation effects with catalysts could be caused by differences in reaction mechanisms. For example, if the RDS involves an electron-transfer process, the cation effect is probably determined primarily by the strength of the electrostatic interactions of cations with the substrate and the activated complexes. Alternatively, cation acidity, that is, the pK_a of the hydrated cation (Table 2), will probably be a relevant variable if the RDS involves a proton-transfer or PCET process. In addition, cation effects show an unexpected crystal face dependence with the same RDS, as shown by facet-specific variations in activity^{16,17}; this implies that cations may have highly structure-sensitive interactions with the electrode surface, leading to inconsistent cation effects on the same reaction catalysed by different crystal facets. The absence of in situ characterization techniques with sufficient spatial resolution has hindered the direct observation of cation distributions near heterogeneous electrode structures. Unravelling the potential synergistic effects between surface motifs and cations in modulating CO₂RR activity requires continued research efforts and methodological innovations.

Conclusions

In this Review we categorize cation effects into three broad groups based on their degree of involvement in CO₂RR (Fig. 1): as indirect modulators in the microenvironment, as energetic regulators of the RDS and as participants in elementary reactions. Although the majority of the existing literature attribute electrostatic interaction and the associated change in the solvent structure as the primary source of the cation effect, recent reports of cations as co-catalysts have introduced

a new perspective for understanding the role of cations in electrocatalytic reactions. Promising future research directions have been outlined to advance the mechanistic understanding of interfacial chemistry and enable the rational design of optimized CO₂RR systems.

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Author contributions

Y.X. and B.X. conceived the work and wrote the initial draft. Y.X., K.Z., X.C. and B.X. contributed to the editing of the manuscript.

Competing interests

The authors declare no competing interests.

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